

1 Easy

Problem 1 (Source: ???)

Let S , A , and M be digits such that

$$17 \times 27 \times 37 \times 47 \times 57 \times 67 = \overline{3,048,SAM,619}.$$

Find $S \times A \times M$.

Solution. Solution



2 Medium-Easy

Problem 2 (Source: ???)

Let $a > 1$ be a real number such that

$$P = a^3 + \frac{1}{a^3} \quad \text{and} \quad Q = a^2 + \frac{1}{a^2}.$$

Given that P and Q are integers, compute the minimum value of PQ .

Solution. Solution

□

3 Medium

Problem 3 (Source: ???)

Find the sum of all not necessarily distinct prime divisors of the sum

$$\sum_{n=1}^{20} \left\lfloor \frac{2^n}{3} \right\rfloor.$$

Solution. Solution

□

4 Hard-Medium

Problem 4 (Source: ???)

Real numbers a, b, c, d satisfy $ab = cd$ and

$$a^2 + b^2 + c^2 + d^2 = 1.$$

Compute the greatest possible value of $29a + 11b + 61c + 19d$.

Solution. Solution



5 Hard

Problem 5 (Source: ???)

Find the largest positive integer $n > 1$ such that $n^2 + \varphi(n)$ divides $n^3 + 1000$. (Recall that $\varphi(m)$ represents the number of positive integers less than m which are relatively prime to m .)

Solution. Solution



6 Hard-Extreme

Problem 6 (Source: ???)

Let $P(x) = 2(x - 1)^2$. Suppose that there are N values of $x \geq \frac{1}{2}$ such that

$$\underbrace{P(P(\cdots P(x) \cdots))}_{2025 \text{ } P's} = x.$$

Find the number of ones in the binary representation of N .

Solution. Solution

□